

L12: Hierarchical Graph Model

The previous models are "stylized" -- in the sense that they have only 1-2 parameters and their main goal is to show how a certain network property (e.g., a power-law degree distribution) can be achieved through a simple mechanism (e.g., preferential attachment).

What if we want to create a network model that meets many of the properties of real-world networks, including: a) heavy-tailed degree distribution, b) strong clustering, c) presence of communities, and d) hierarchical structure (small and tight communities embedded in larger but looser communities)?

We will now present a model that can meet, to some extent, all these properties. The model is referred to as *Hierarchical Random Graph (HRG)* and it was proposed by Aaron Clauset and his colleagues.

As opposed to the previous models in this lesson, HRG requires many parameters. Further, these parameters can be calculated algorithmically based on network data, so that the resulting model matches closely the hierarchical density structure of the given network.

First, let us describe the HRG model -- without considering the parameter estimation problem. We will examine that problem in the next few pages.

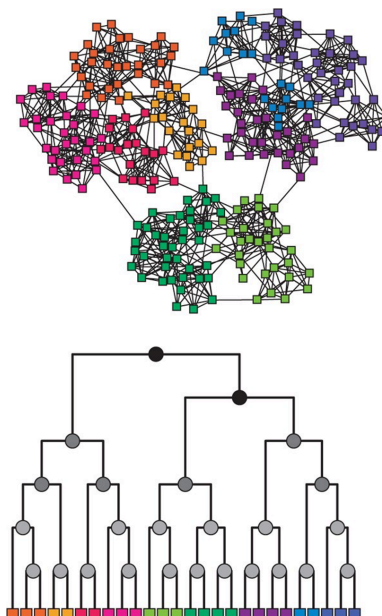
So, suppose we want to create a graph G with n nodes, using the HRG model. The model is described by a *dendrogram* D , which is a binary tree with n leaves (and so the number of internal nodes in D is $n-1$).

Each internal node r is associated with a probability p_r -- this is the probability that two nodes are connected if r is their lowest common ancestor in D . So, if two graph nodes i and j have the same parent r in D , the two nodes are connected with probability p_r .

On the other extreme, if r is the root of D , the probability p_r represents the probability that any node of G at the left subtree of r connects with any node at the right subtree of r .

So, the complete description of the HRG model requires:

1. The specification of a dendrogram D with n leaves
2. An $(n-1)$ -dimensional vector of probabilities p_r .



The HRG model is very flexible and it can encompass many different graph models. For instance, if all probabilities p_r are equal to the same value p , we get the $G(n,p)$ model. On the other hand, if we want to create a graph with a strong community structure (as shown in the visualization) we can set p_r to a high value for all internal nodes in D that are *lowest common ancestors (LCA)* of nodes in the same community.

For example, consider the green community in the visualization – the internal nodes that are LCAs to only green nodes can have a high value of p_r – at least relative to the value of p_r for nodes that are LCAs of nodes that belong to different communities. Similarly for the other communities.

Additionally, if we want to create hierarchical communities, we can have that the lower internal nodes in D have higher values of p_r than the higher nodes in D . This pattern can also generate strong local clustering.

Food For Thought

- a) Can the HRG model generate networks with degree correlations? (assortative or disassortative)
- b) How would you set the vector of p_r probabilities to create a skewed degree distribution?